DBMS

Data – raw facts and figures
Information – processed data

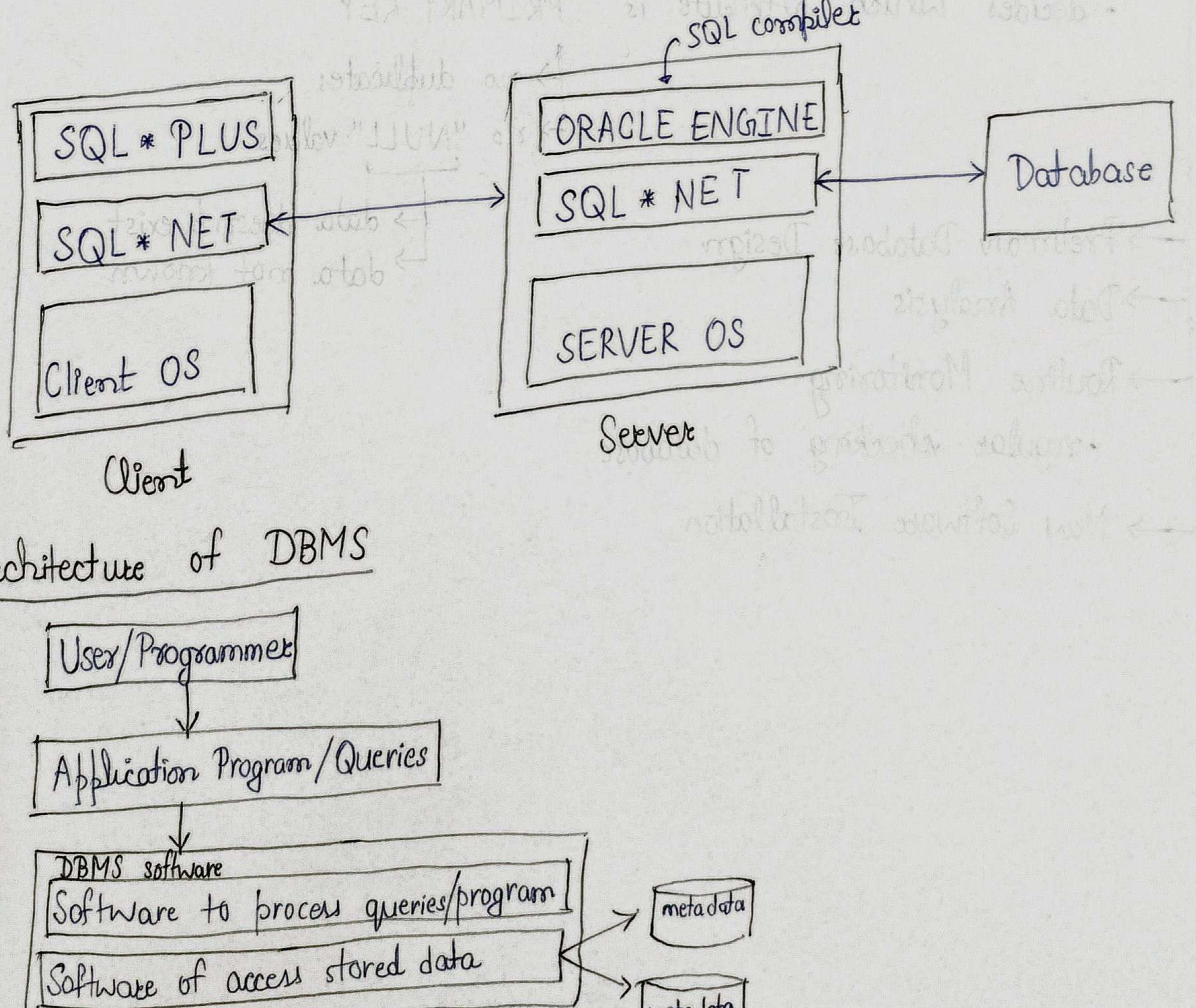
Database – collection of related data from which user can retrieve the desired data easily

. Data stored in form of tables.

* Tuple – row of table
* Attribute – column of table

DBMS – software through which
. database is created
. data is modified in the database
. data is retrieved from the database

e.g.- ORACLE



Architecture of DBMS

Users of DBMS

(I) End Users
- → Naive Users — interacts using some application software
- → Sophisticated Users — interacts by writing their own queries

(II) System Analysts

(III) Application Programmer

(IV) DBA (Database Administrator)

→ Schema definition and modification

Schema (structure of database)

Schema – overall structure of database
Instance – at a particular time, the state of data stored in the database

→ Security enforcement and administration
- controls who can access what portion of the database
- decides constraints on the database
- decides which attribute is "PRIMARY KEY"
  - → no duplicates
  - → no "NULL" values
    - → data doesn't exist
    - → data not known

→ Preliminary Database Design

→ Data Analysis

→ Routine Monitoring
- regular checking of database

→ New Software Installation

DBMS v/s Traditional File system

(i) Controlling Redundancy
- duplication of data
  - → wastage of storage space
  - → inconsistency among different instances of the same data

* Normalization – technique used in DBMS to avoid inconsistency by dividing larger schemas into smaller ones

(ii) Restriction of unauthorized access

(iii) Providing search technique for efficient query process (Query Optimization)

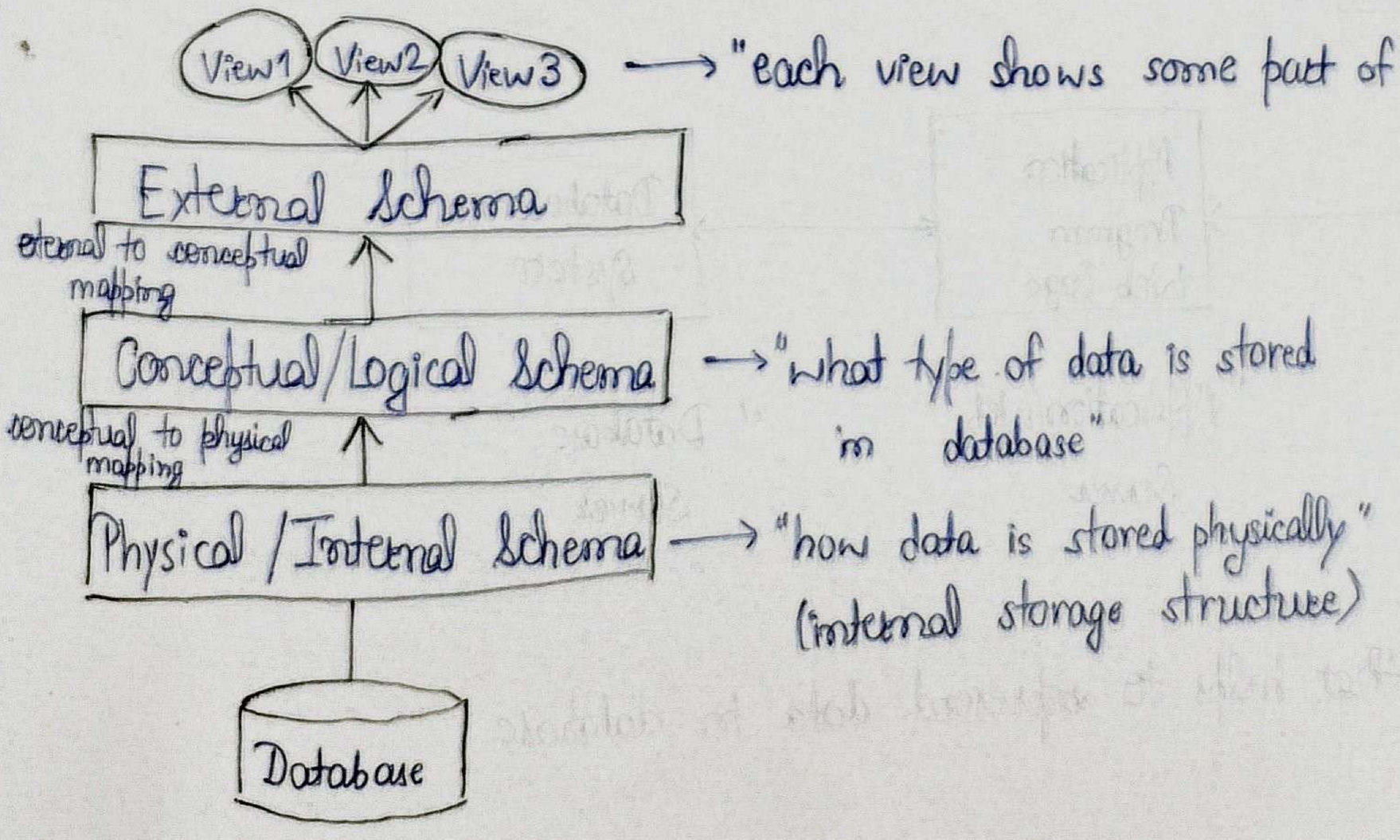
(iv) Backup and Recovery

(v) Enforcing Integrity Constraint
- → Entity Integrity Constraint (primary key constraint)
- → Referential Integrity Constraint (foreign key constraint)
- → Check Constraint (checks for a specific condition)

(vi) Permitted Interference and Access Rule

(vii) Providing multiple user interface

Three Schema Architecture



→ "each view shows some part of database"

→ "what type of data is stored in database"

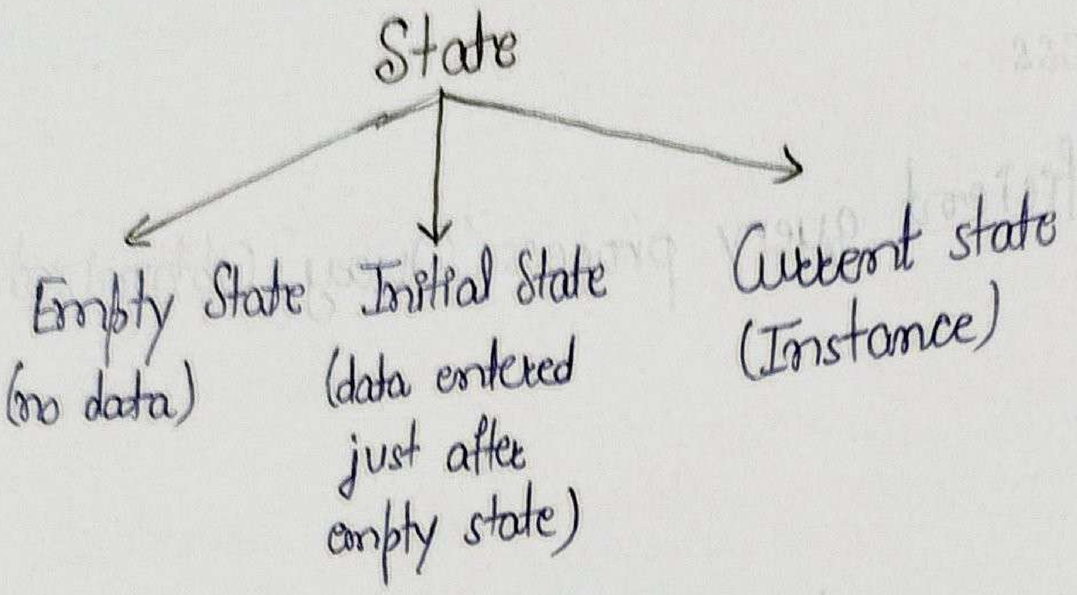
→ "how data is stored physically" (internal storage structure)

Data Types in SQL:

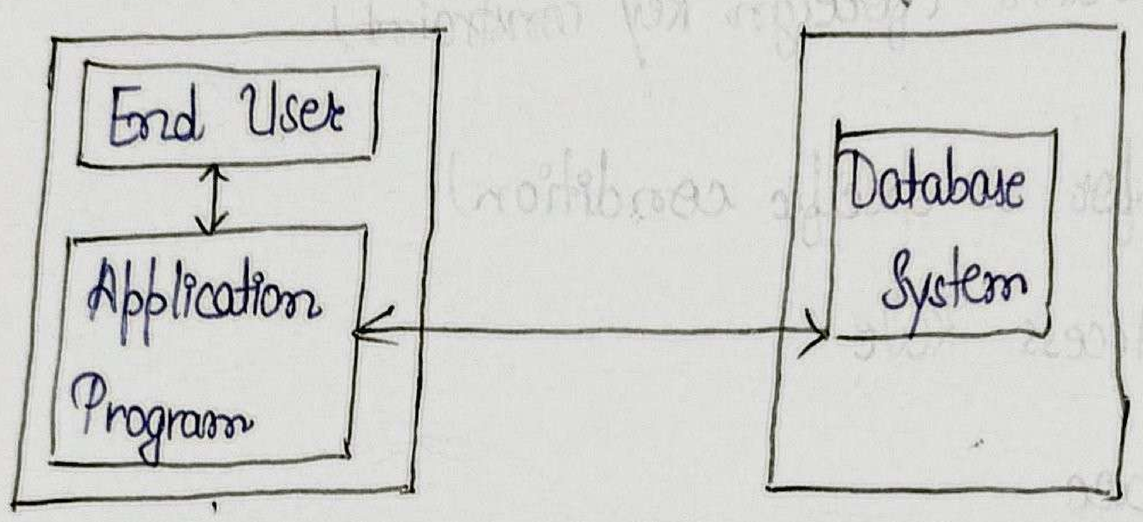
char(n), varchar(n), varchar2(n), number(n)

Data Independence

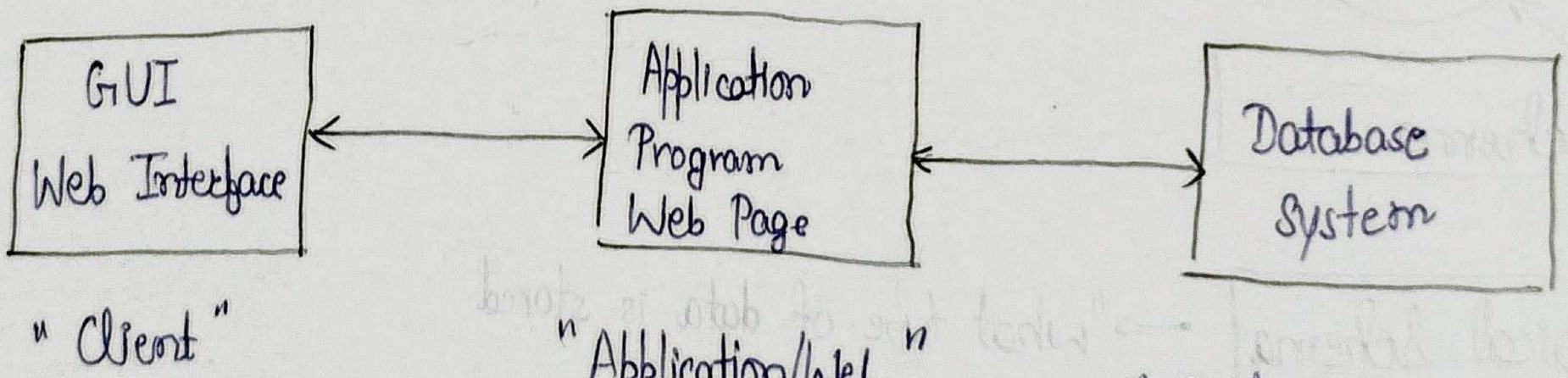
- Physical Data Independence — changing physical structure doesn't affect the logical layer
- Logical Data Independence — conceptual structure changes doesn't affect the view level



Two tier Architecture



Three Tier Architecture



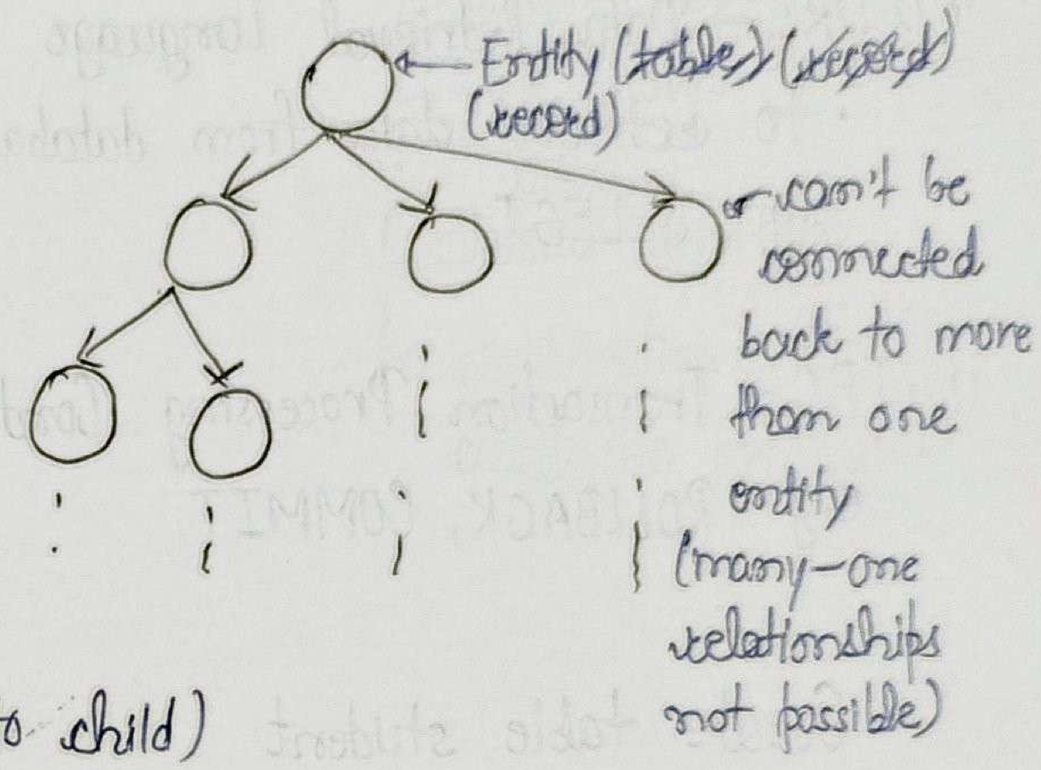
* Data Model — tool that helps to represent data in database

Data Model

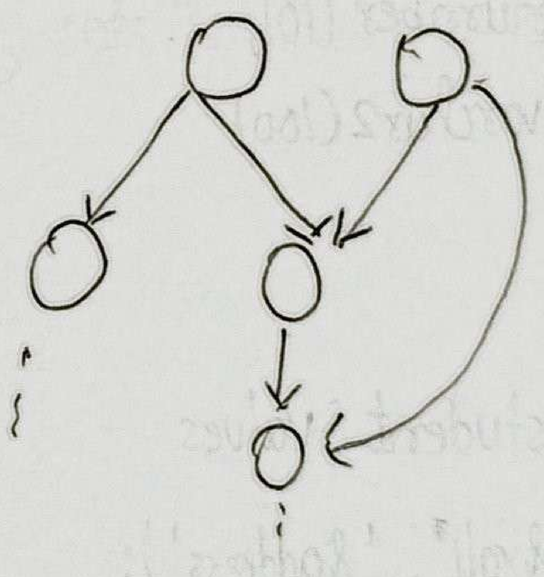
- Object-based Model (Object Oriented Model / Entity Relationship (ER) Model)
  - used at external and conceptual levels
  - uses entities (objects) and relationships b/w them
    - in ER model
    - in object-oriented model
- Physical Model → used at physical level
- Record-based Model
  - each tuple treated as a "record"

Record-based Model

- Hierarchical Model
  - 1:n relationship
  - oldest data model
  - easy to implement
  - lack of structural independence
  - operational anamoly (entities need to traversed from parent to child)



- Network Model
  - one child can be connected to more than one parent
  - implementation is complex
  - more structural independence



- Relational Model
  - data is stored in "tables"
  - rows – tuples
  - columns – attributes
  - * Degree of Relation: no. of attributes/columns
  - * Cardinality of Relation: no. of tuples/rows
  - structural independent
  - conceptually simpler.
  - easy to design and maintain

Database Language

(i) DDL – Data Definition Language
- to define the structure of data in database

e.g. - CREATE

(ii) DML – Data Manipulation Language
- to modify data in database

e.g. - UPDATE, DELETE, INSERT, ALTER

(iii) DRL – Data Retrieval Language
- to retrieve data from database

e.g. - SELECT

(iv) TPC – Transaction Processing Control

e.g. - ROLLBACK, COMMIT

```
Create table student
( name    varchar2(10),
  roll    number(10),
  address varchar2(100)
);
```

```
Insert into student values
('&name', '&roll', '&address');
```

Entity - Relationship (ER) Model (Object-based Model)

* Entity — any real life object
* Entity Set — set of similar entities

Attribute — property through which entity can be identified

(I) simple → indivisible

(II) composite → divisible into smaller parts

(I) single-valued → only has one value

(II) multi-valued → can have multiple values

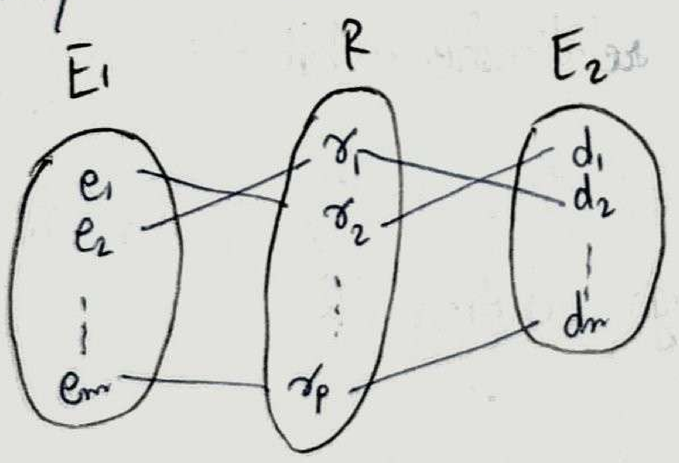
(I) store → attribute from which other attributes can be derived

(II) derived → attribute found from other attributes

* NULL attribute
  - ↳ doesn't exist
  - ↳ not known to DB designer

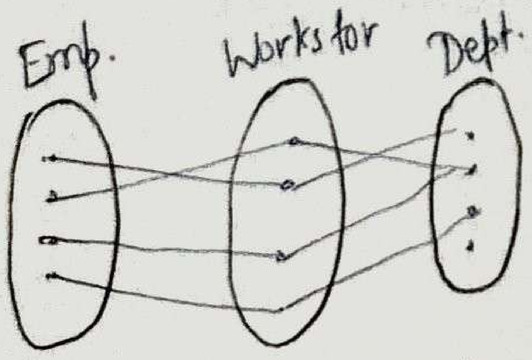
* ~~Composite~~ Complex attribute
  - ↳ combination of multiple composite attributes

* Relation: A relationship 'R' between two entity sets 'E_1' and 'E_2' associates one entity of 'E_1' with a entity of 'E_2'

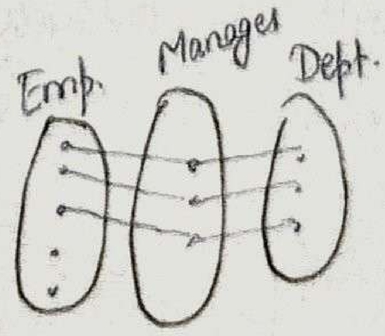


· Full Participation — each entity of E_1 is connected to some entity of E_2 through the relation 'R'.

· Partial Participation — not all entities of E_1 are connected to some entity of E_2 through the relation 'R'



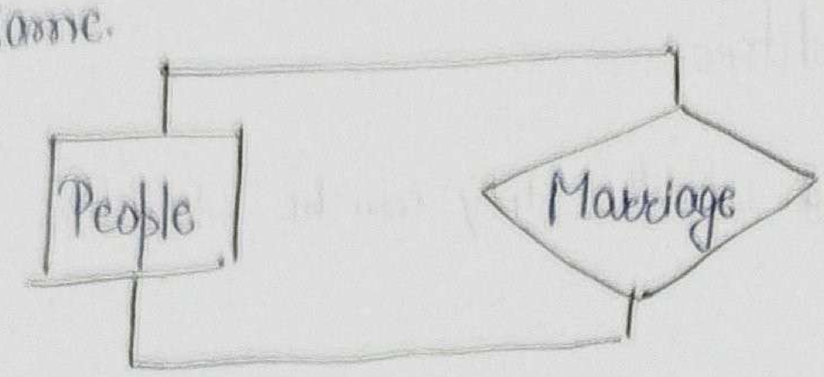
Full Participation



Partial Participation

* Degree of Relation: no. of entity sets connected through that relation.

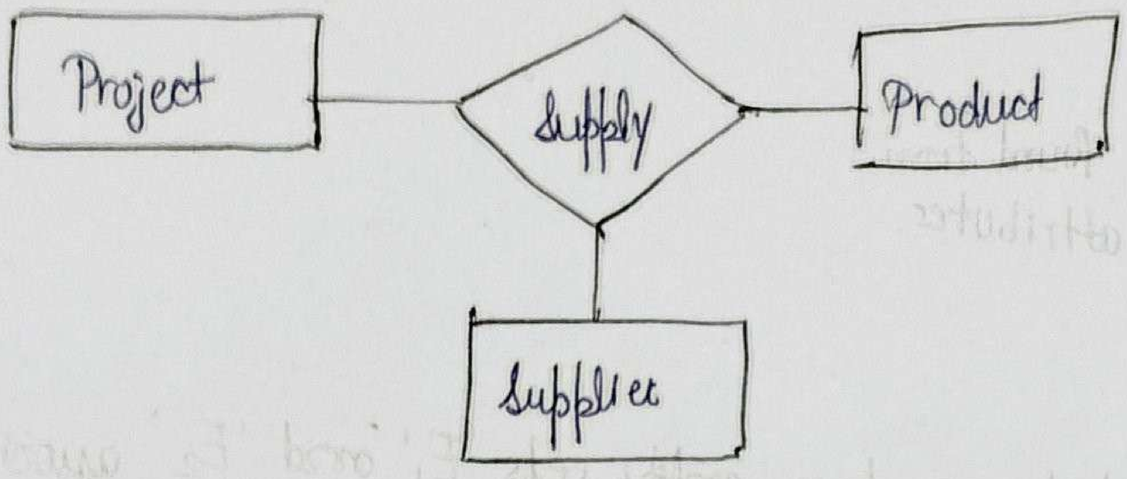
(I) Unary Relation — associates one entity of an entity set to another entity of the same.



(II) Binary Relation — associates one entity of an entity-set 'E_1' to an entity of another entity set 'E_2'



(III) Ternary Relation — associates entities between three entity sets.



* Domain of an attribute: range of attributes acceptable for that particular attribute

* Role of a relation: name of the relationship (what it is doing)

* Cardinality Ratio: no. of entities connected from entity set 'E_1' to no. of entities connected to that of entity set 'E_2' through relation 'R'

  - (I) 1:1 (one-one)
  - (II) M:N (many-many)
  - (III) M:1 (many-one)
  - (IV) 1:M (one-many)

* Weak Entity Set: can't be uniquely identified by its own attributes alone, and depends on a related strong entity set for identification (no PRIMARY KEY)

